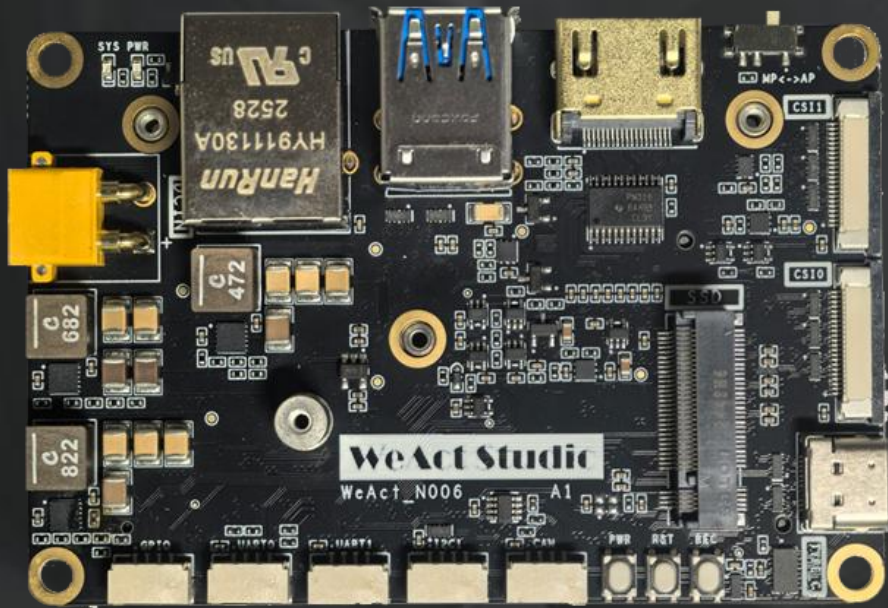




WeAct Studio

N006

CARRIER BOARD

Tutorial

WEACT Studio

Catalog

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REVISION HISTORY

Draft Date	Revision	Description
2026.02.24	V1.0	1. Initial
2026.06.07	V1.1	1. Support Jetpack7.x

1. BUILD FLASH ENVIRONMENT

- a) First, you need a computer with **Ubuntu 20.04** or above as the host to flash Orin NX, or you can install VMware on windows.

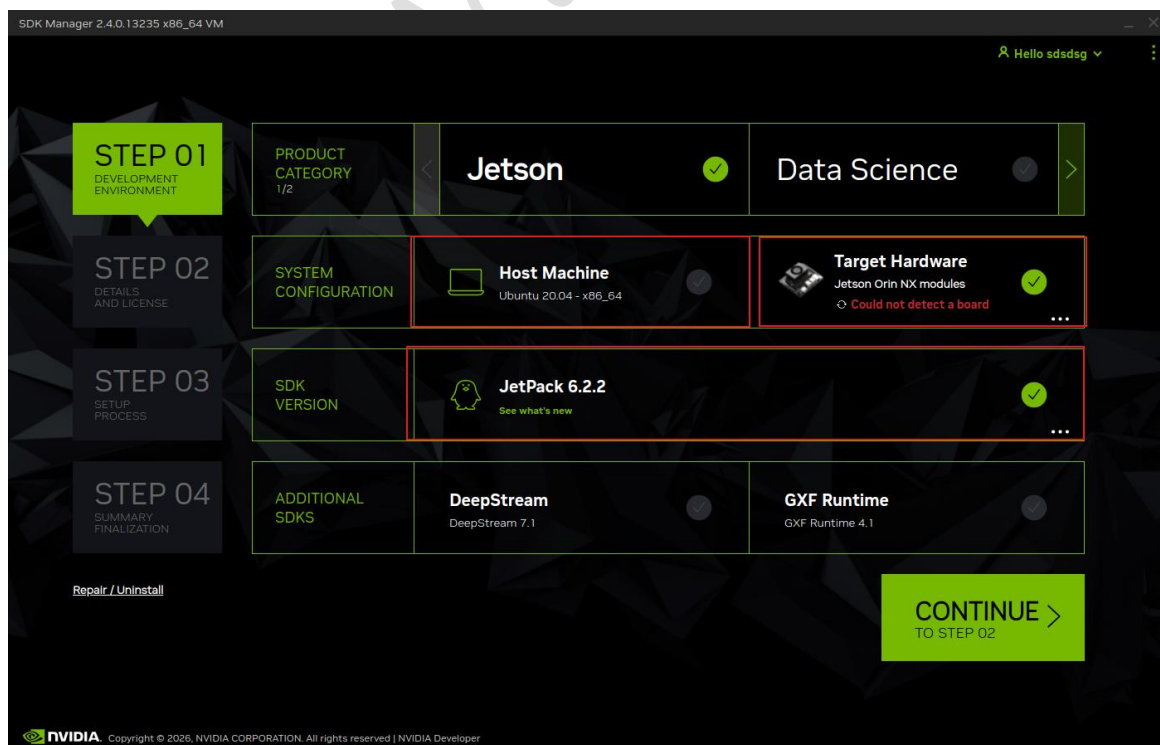
****To flash Jetpack 7.x, an Ubuntu 24.04 computer is required as the HOST**

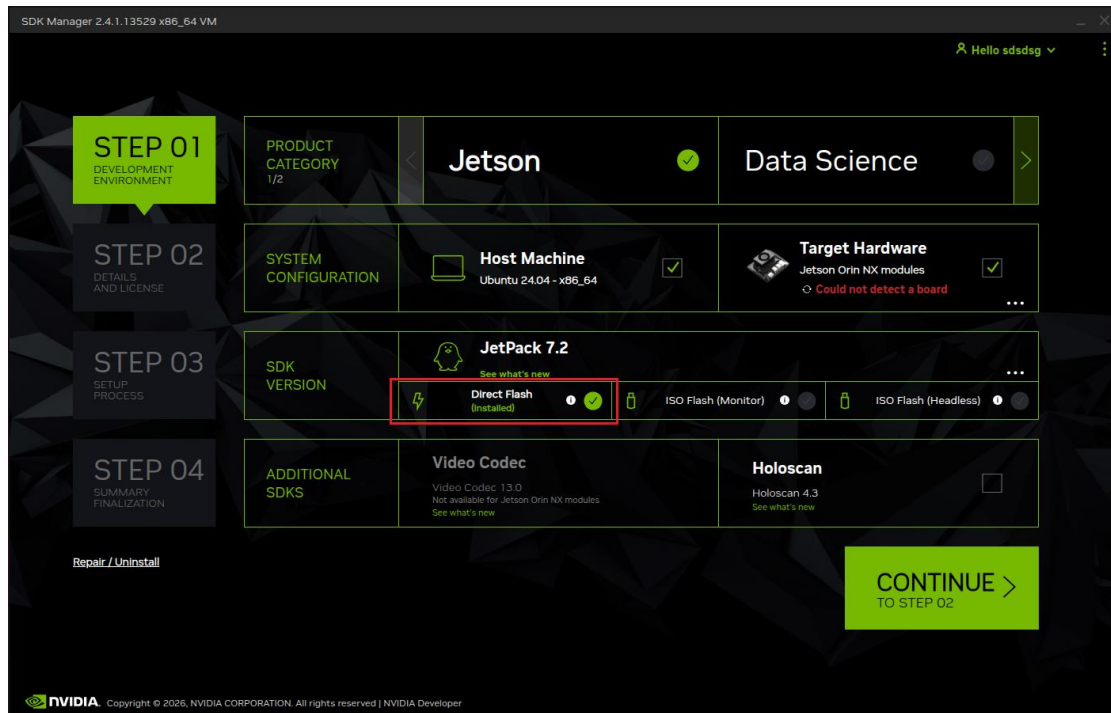
- b) Download the latest **SDK manager** from NVIDIA and install it in **Ubuntu 20.04** (You need to register an NVIDIA account, which will also be used later)

➤ SDK-Manager Download: <https://developer.nvidia.com/nvidia-sdk-manager>

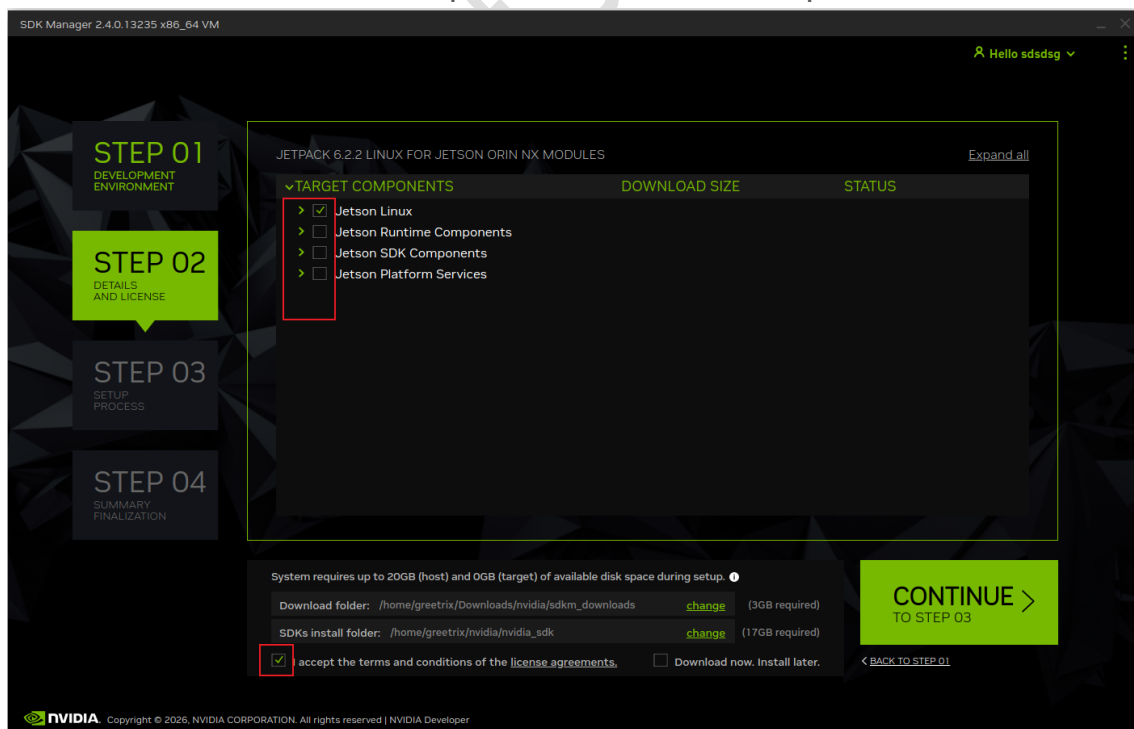
- c) **Continue** Select the required **Target Hardware** and **JetPack** version, **uncheck Host Machine (to save host storage space)**, and check DeepStream/GXF Runtime based on component requirements. Here, **Jetpack 6.2.2 version is selected as an example**, then click **Continue**

***For Jetpack 7.x, Select Direct Flash**





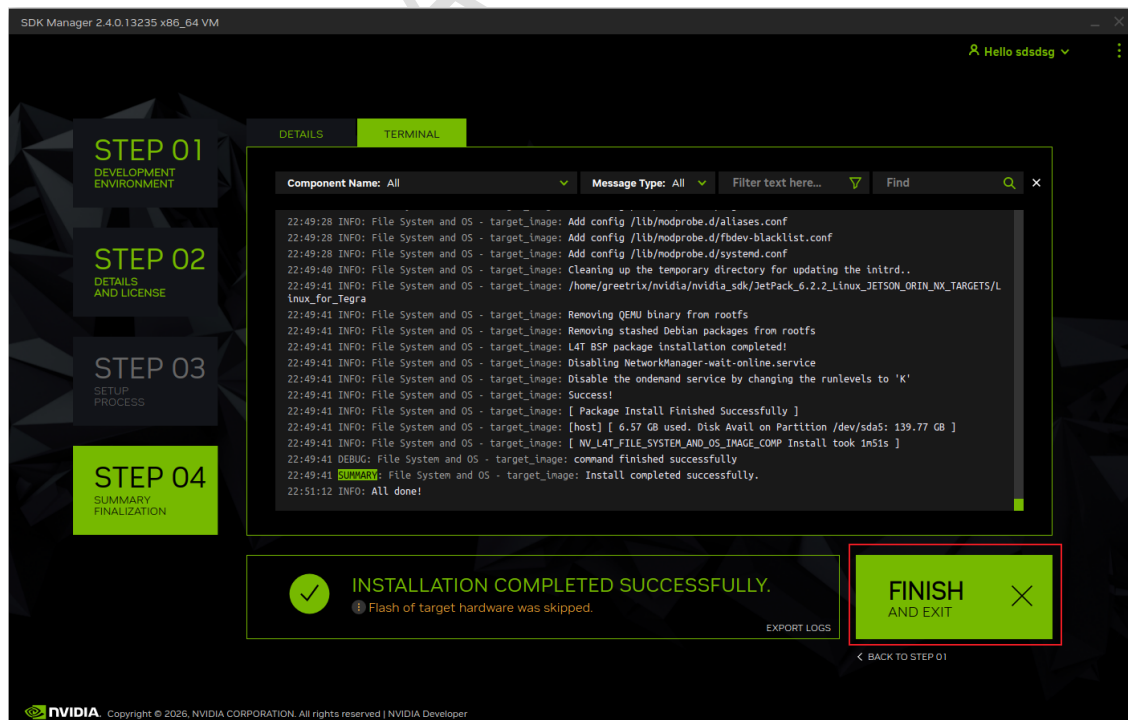
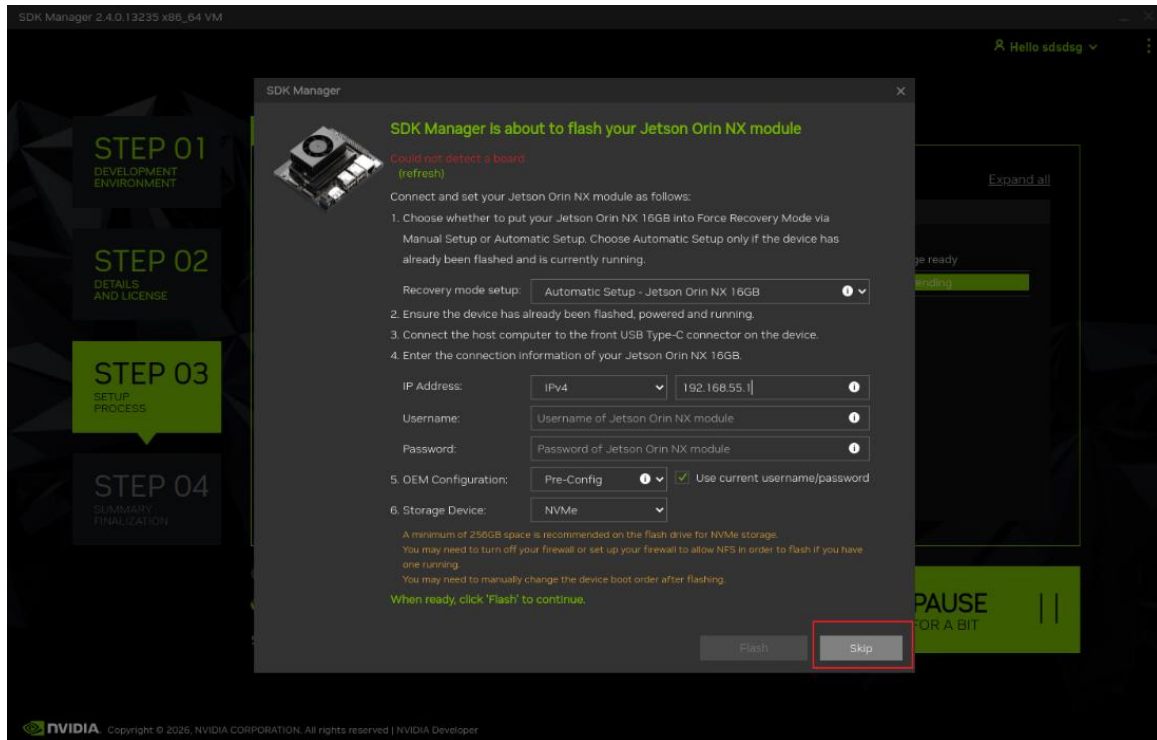
- d) Only select **Jetson Linux** here, **uncheck Jetson Runtime/SDK Components/Platform Services** (the components will be re downloaded and installed together later), and click **CONTINUE** to proceed to the next step.



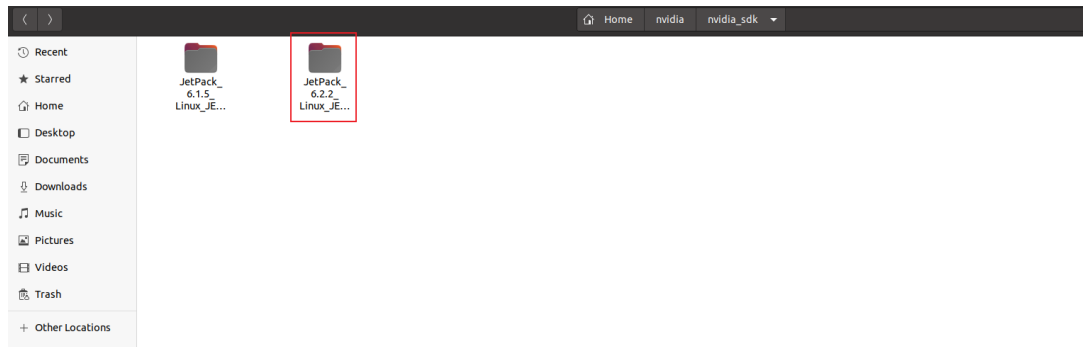
P. S: Please download and install in a smooth network environment. When the download or installation fails, click **Retry** to continue until all the status is installed

and green is displayed. During the installation process, a network burning message will pop up and select skip.

(The subsequent steps involve automatically replacing the device tree and flashing the system via scripts)

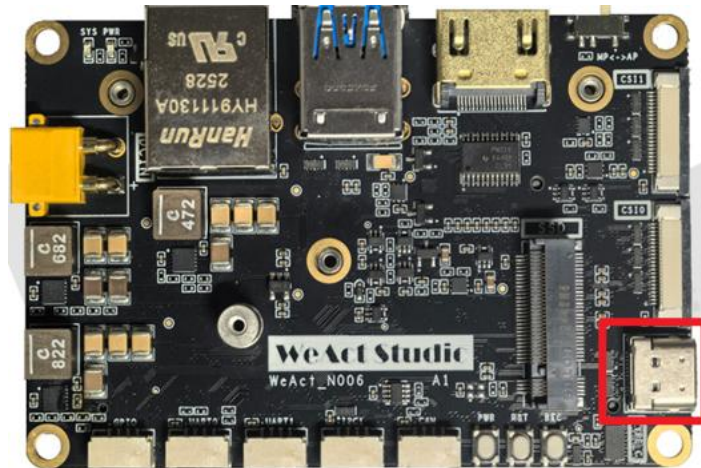


- e) After the installation is successful, the required files will be downloaded with the corresponding version under `~/NVIDIA/nvidia_sdk/`

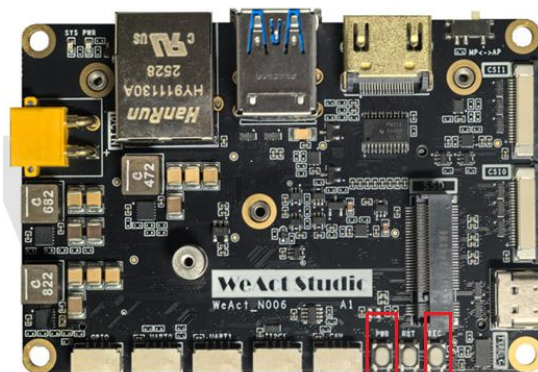


2. FLASH SYSTEM IMAGE

1. Connect **USB Type-C** Cable to carrier board **USB OTG** connector



1. Switch slide button to MP(**Manual Power On**), 2 way enter Recovery mode below
 - a) Long press **REC** key, Press **PWR** to power on, Release **REC** key that make the system to enter Recovery mode
 - b) Press **PWR** to power on, Long Press **REC** Key, Press **RST** to enter Recovery mode
2. **NVIDIA USB Driver** will appear in the lower right of VMware, or open the terminal and enter **lsusb** command, the NVIDIA Corp will be found and the fan speed will be set to max.



PWR

REC

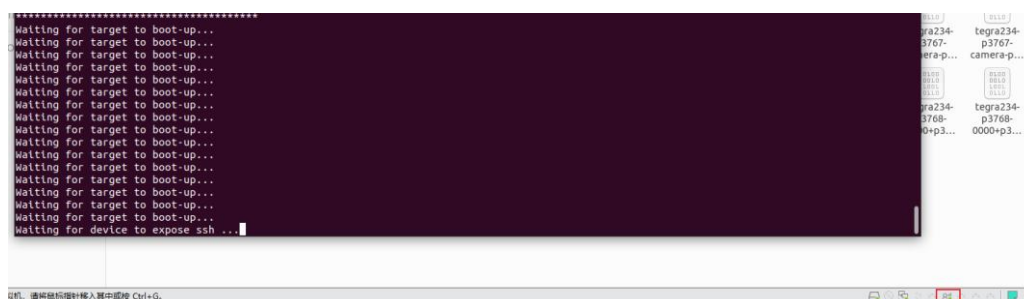

```
[INFO] Script directory: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script
[INFO] Current directory: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
[SUCCESS] Found Linux_for_Tegra directory: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
[INFO] Detected JetPack version: 6.2.2
[INFO] Major version: 6
[INFO] Copying files for JetPack 6
[INFO] Copied Kernel DTB: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/t
egra234-p3768-0000+p3767-0000-nv.dtb -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel/dtb/
[INFO] Copied Kernel DTB: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/t
egra234-p3768-0000+p3767-0000-nv-super.dtb -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel
/dtb/
[INFO] Copied Kernel DTB: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/t
egra234-p3768-0000+p3767-0001-nv.dtb -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel/dtb/
[INFO] Copied Kernel DTB: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/KN_DTB/t
egra234-p3768-0000+p3767-0001-nv-super.dtb -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/kernel
/dtb/
[INFO] Copied GPIO BCT: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb1-bct-padvoltage-p3767-dp-a03.dtsi -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/boot
loader/generic/BCT/
[INFO] Copied GPIO BCT: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb1-bct-padvoltage-p3767-hdmi-a03.dtsi -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/boot
loader/generic/BCT/
[INFO] Copied GPIO BCT: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb1-bct-pinnux-p3767-dp-a03.dtsi -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootload
er/generic/BCT/
[INFO] Copied GPIO BCT: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb1-bct-pinnux-p3767-hdmi-a03.dtsi -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloa
der/generic/BCT/
[INFO] Copied GPIO BCT: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BCT/t
egra234-mb2-bct-misc-p3767-0000.dts -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader/ge
neric/BCT/
[INFO] Copied GPIO Info: /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/script/DTB/JP6/GPIO_BL/t
egra234-mb1-bct-gpio-p3767-dp-a03.dtsi -> /home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra/bootloader
```

Script will replace the device-tree and flash the system auto.

```
greatrix@ubuntu: ~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
[ 0.1178 ] Added binary blob tsec_t234_sigheader.bin.encrypt of size 192512
[ 0.1200 ] Added binary blob applet_t234_sigheader.bin.encrypt of size 279808
[ 0.1218 ] Not supported type: mb2_applet
[ 0.1228 ] Added binary blob mb2_t234_with_mb2_cold_boot_bct_MB2_sigheader.bin.encrypt of size 440880
[ 0.1243 ] Added binary blob xusb_t234_prod_sigheader.bin.encrypt of size 164864
[ 0.1276 ] Added binary blob nvpsa_020_sigheader.fw.encrypt of size 2164640
[ 0.1280 ] Added binary blob display_t234-dce_sigheader.bin.encrypt of size 12078816
[ 0.1285 ] Added binary blob nvdec_t234_prod_sigheader.fw.encrypt of size 294912
[ 0.1288 ] Added binary blob bmpm_t234-TE980M-A1_prod_sigheader.bin.encrypt of size 1027072
[ 0.1299 ] Added binary blob tegra234-bmpm-3767-0000-3768-super_with_odm_sigheader.dtb.encrypt of size 383296
[ 0.1305 ] Added binary blob camera-rtcpu-t234-rce_sigheader.img.encrypt of size 458096
[ 0.1309 ] Added binary blob adsp-fw_sigheader.bin.encrypt of size 124832
[ 0.1313 ] Added binary blob spe_t234_sigheader.bin.encrypt of size 270336
[ 0.1317 ] Added binary blob tos-optec_t234_sigheader.img.encrypt of size 1932448
[ 0.1321 ] Added binary blob eks_t234_sigheader.img.encrypt of size 9232
[ 0.1326 ] Added binary blob boot0.img of size 80502784
[ 0.1494 ] Added binary blob tegra234-p3768-0000+p3767-0000-nv-super.dtb of size 249565
[ 0.2023 ] tegrarcv_v2 --instance 3-2 --chip 0x23 0 --pollbl --download bct_mem mem_rcm_sigheader.bct.encrypt --download blob blob.bin
[ 0.2028 ] BL: version 1.4.0-7-t234-54845784-56c609fa last_boot_error: 0
[ 0.3243 ] Sending bct_mem
[ 0.3285 ] Sending blob
[ 4.4299 ] RCM-boot started

/home/greatrix/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
*****
* Step 3: Start the flashing process *
*****
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
Waiting for target to boot-up...
```

Not USB will re-connect. If it is a virtual machine, please grant the permission for automatic USB connection in advance

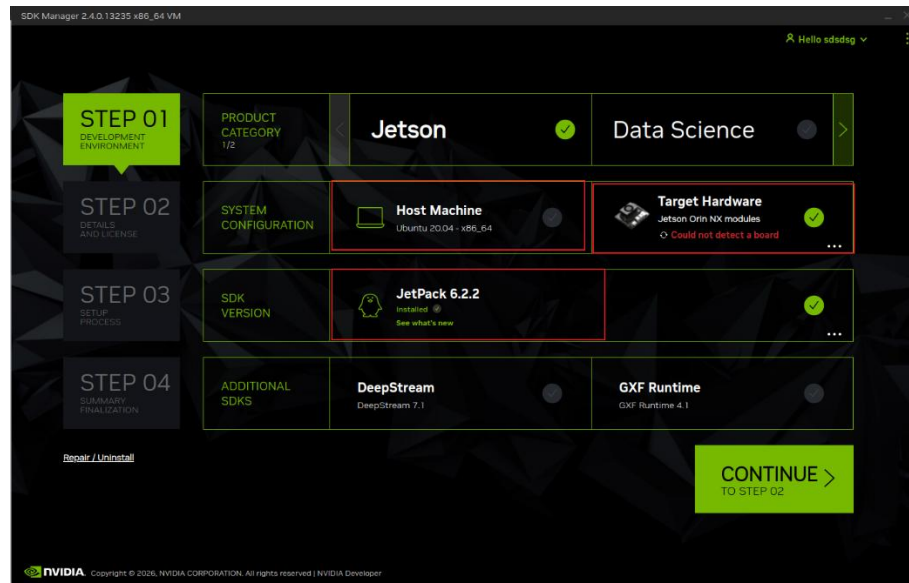


```
greetrix@ubuntu: ~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra
00:03:48.528 - Warning: Skip writing uefi_variables partition as no image is specified
00:03:48.551 - Warning: Skip writing uefi_ftw partition as no image is specified
00:03:48.574 - Warning: Skip writing reserved partition as no image is specified
00:03:48.597 - Warning: Skip writing worn partition as no image is specified
00:03:48.621 - Debug: Writing bct_backup.img (partition: BCT-boot-chain_backup) into /dev/mtdd0
00:03:48.629 - Debug: Sha1 checksum matched for /mnt/internal/bct_backup.img
00:03:48.633 - Debug: Writing /mnt/internal/bct_backup.img (32768 bytes) into /dev/mtdd0:66715648
Copied 32768 bytes from /mnt/internal/bct_backup.img to address 0x03fa0000 in flash
00:03:48.707 - Warning: Skip writing reserved partition as no image is specified
00:03:48.730 - Debug: Writing gpt_backup_secondary_3_0.bin (partition: secondary_gpt_backup) into /dev/mtdd0
00:03:48.738 - Debug: Sha1 checksum matched for /mnt/internal/gpt_backup_secondary_3_0.bin
00:03:48.741 - Debug: Writing /mnt/internal/gpt_backup_secondary_3_0.bin (16896 bytes) into /dev/mtdd0:66846720
Copied 16896 bytes from /mnt/internal/gpt_backup_secondary_3_0.bin to address 0x03fc0000 in flash
00:03:48.791 - Debug: Writing qspt_bootblob_ver.txt (partition: B_VER) into /dev/mtdd0
00:03:48.798 - Debug: Sha1 checksum matched for /mnt/internal/qspt_bootblob_ver.txt
00:03:48.801 - Debug: Writing /mnt/internal/qspt_bootblob_ver.txt (109 bytes) into /dev/mtdd0:66912256
Copied 109 bytes from /mnt/internal/qspt_bootblob_ver.txt to address 0x03fd0000 in flash
00:03:48.825 - Debug: Writing qspt_bootblob_ver.txt (partition: A_VER) into /dev/mtdd0
00:03:48.831 - Debug: Sha1 checksum matched for /mnt/internal/qspt_bootblob_ver.txt
00:03:48.834 - Debug: Writing /mnt/internal/qspt_bootblob_ver.txt (109 bytes) into /dev/mtdd0:66977792
Copied 109 bytes from /mnt/internal/qspt_bootblob_ver.txt to address 0x03fe0000 in flash
00:03:48.858 - Debug: Writing gpt_secondary_3_0.bin (partition: secondary_gpt) into /dev/mtdd0
00:03:48.866 - Debug: Sha1 checksum matched for /mnt/internal/gpt_secondary_3_0.bin
00:03:48.870 - Debug: Writing /mnt/internal/gpt_secondary_3_0.bin (16896 bytes) into /dev/mtdd0:67091968
Copied 16896 bytes from /mnt/internal/gpt_secondary_3_0.bin to address 0x03ffe000 in flash
00:03:48.901 - Info: Successfully flashed the QSPI.
00:03:48.905 - Info: Flashing success
00:03:48.912 - Debug: The device size indicated in the partition layout xml is smaller than the actual size. This utility will try to fix the GP
Flash is successful
Reboot device
Cleaning up...
Log is saved to Linux_for_Tegra/initrdlog/flash-3-2-0_20260208-225849.log
[SUCCESS] Flash done. Please wait the Jetson Orin boot.
greetrix@ubuntu:~/nvidia/nvidia_sdk/JetPack_6.2.2_Linux_JETSON_ORIN_NX_TARGETS/Linux_for_Tegra$
```

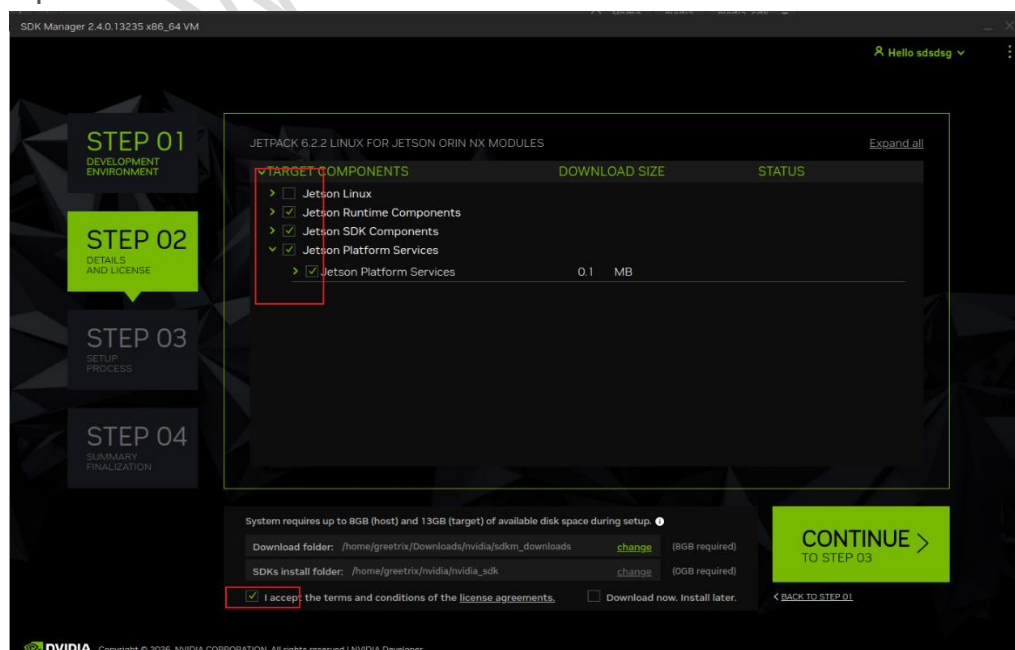
After updating the device tree, it will be successful! Display, as shown in the following figure

3. INSTALL NVIDIA COMPONENT

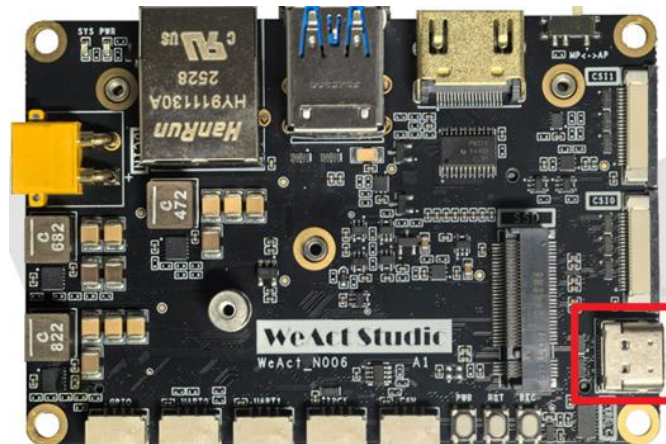
1. Select the target **hardware** and **jetpack** version required, uncheck the host machine, and click **CONTINUE**.



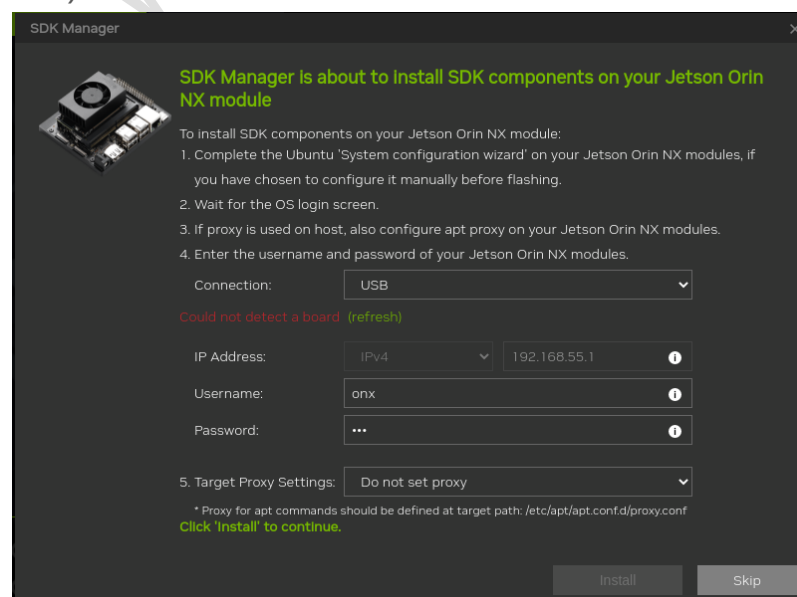
2. Check the required **SDK components**, check **I accept the terms and conditions of the license agreements**, and click **CONTINUE** to proceed to the next step.

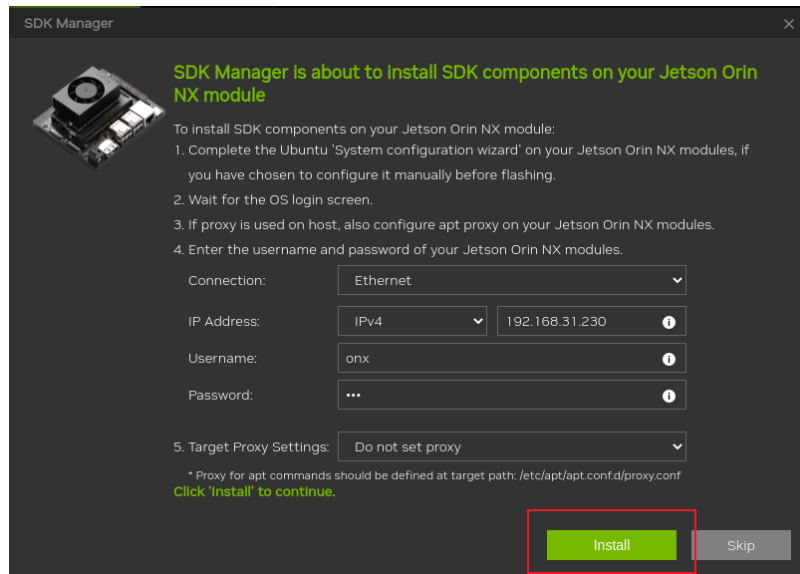


3. Use the **USB type-C cable** to connect the **USB OTG interface** on the carrier board.

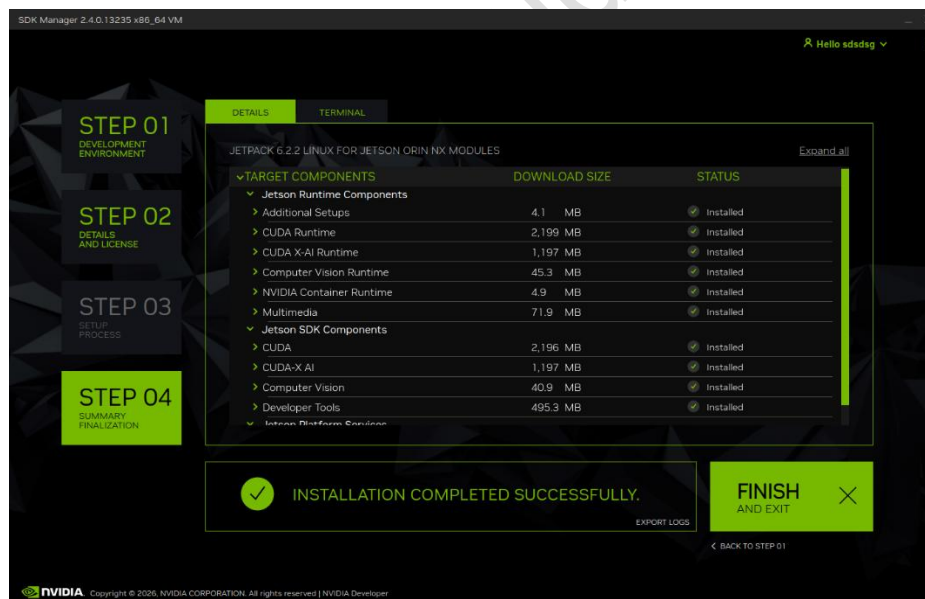


4. Turn the **power on key to MP (manual power on)** and **press PWR key** to power on. At this time, **NVIDIA USB drive sign** will appear in the lower right corner of VMware, or open the terminal and enter **lsusb** command to find **NVIDIA Corp.**
5. Enter the **Orin NX** account password, and ensure the **Orin NX device remains connected to the internet**. Alternatively, you can use the IP address within the same local network. (Figure 2: Host and ORIN NX connected to the same router)



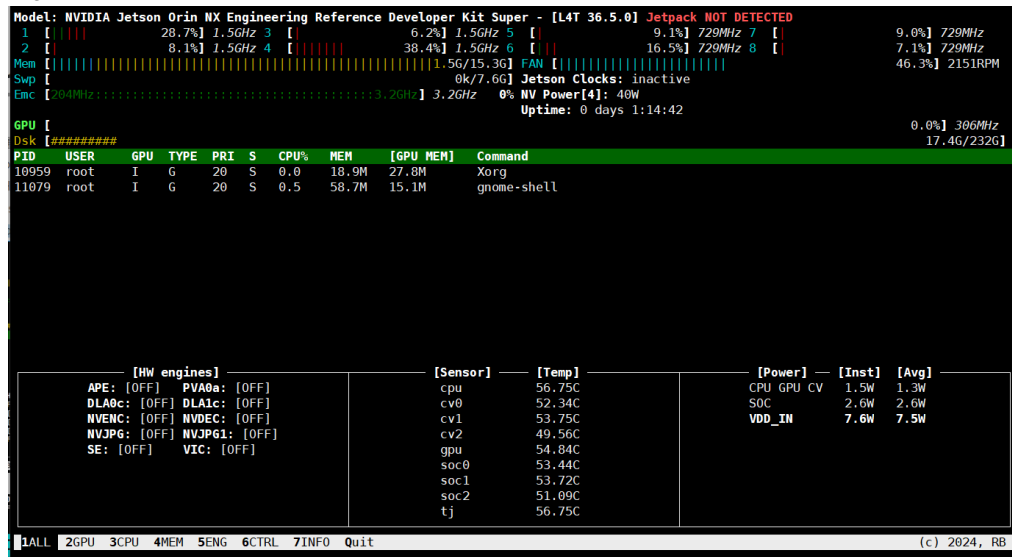


6. Wait for the installation to complete.

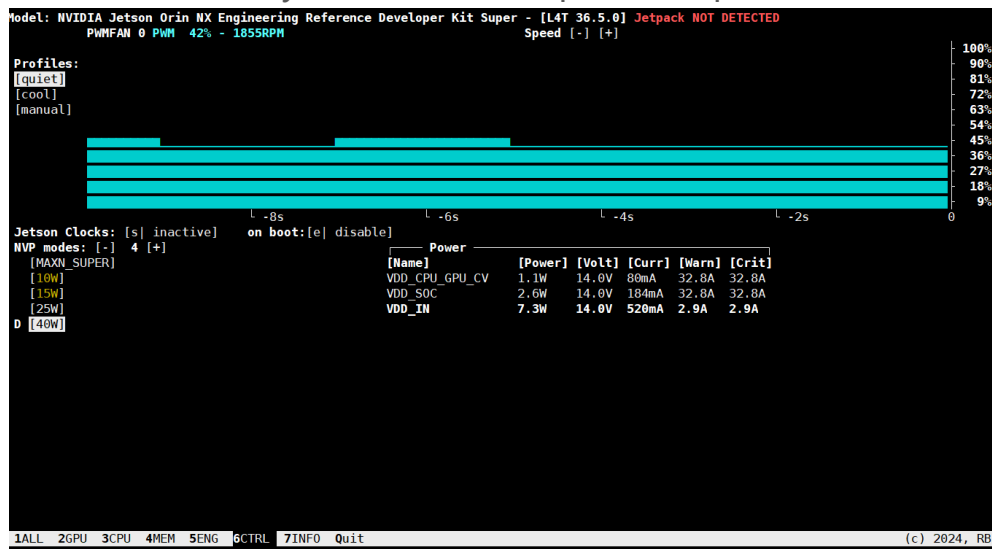


4. INSTALL JTOP(CONTROL FANS/GET SYSTEM INFO)

1. Open terminal, use below command install JTOP
 - a) `sudo apt-get install python3-pip python3-dev -y`
 - b) `sudo -H pip3 install jetson-stats --break-system-packages`
 - c) `sudo systemctl restart jtop.service`
 - d) `sudo jtop`



2. Click on 6CTRL to manually control the fan speed and power scheme.

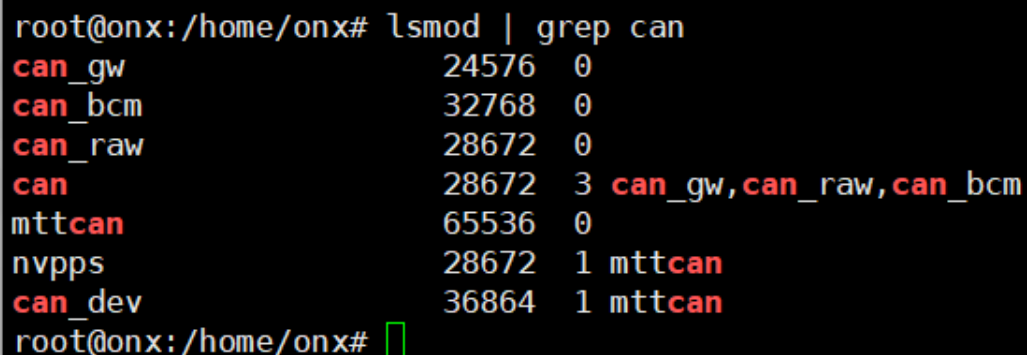


5. USING CAN FOR COMMUNICATION

1. OrinNX integrates one CAN controller CAN0, and WeAct Studio has designed one CAN transceiver (CAN0) on the carrier board, which can **be directly mounted on the CAN physical bus** for use.
2. OrinNX comes with a canbus driver and is integrated into the image, supporting canbus without further processing. We need to install the canbus module. (Enter the following command on the terminal or put it into rc.local to enable self start)

```
sudo modprobe can
sudo modprobe can-raw
sudo modprobe can-bcm
sudo modprobe can-gw
sudo modprobe can_dev
sudo modprobe mttcan
```

3. Check if the installation was successful through **lsmod**.



```
root@onx:/home/onx# lsmod | grep can
can_gw                24576  0
can_bcm               32768  0
can_raw               28672  0
can                   28672  3 can_gw,can_raw,can_bcm
mttcan                65536  0
nvpps                 28672  1 mttcan
can_dev               36864  1 mttcan
root@onx:/home/onx#
```

4. Configure canbus properties, similar to the baud rate setting of the serial port.

```
sudo ip link set can0 type can bitrate 500000
```

```
sudo ip link set up can0
```

5. Check if the configuration was successful through `ifconfig`.

```
root@onx:/home/onx# sudo ip link set can0 type can bitrate 500000
root@onx:/home/onx# sudo ip link set up can0
root@onx:/home/onx# ifconfig
can0: flags=193<UP,RUNNING,NOARP> mtu 16
    unspec 00-00-00-00-00-00-00-00-00-00-00-00-00-00-00-00 txqueuelen 10  (UNSPEC)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
    device interrupt 201
```

- Through cansend can0 (can1) on a terminal `xxx` Command to send data, and another terminal completes the actual signal transmission and reception test through candump can1 (can0)

[illegible]

6. GPIO USAGE IN SHELL

1. Jetson OrinNX can control the GPIO input or output through shell command, install gpiod by below command.
 - a) `sudo apt-get install gpiod`
 - b) `sudo apt-get install libgpiod-dev`

2. GPIO Usage

	Orin NX 8G/16G
GPIO1	PG.06
GPIO2	PAC.06
GPIO3	PN.01
GPIO4	PH.00

- a. GPIO Output: Example with GPIO1
 - i. `gpioset `gpiofind "PG.06"`=0` (Set Output Low Level)
 - ii. `gpioset `gpiofind "PG.06"`=1` (Set Output High Level)
- b. GPIO Input: Example with GPIO1
 - i. `gpioget `gpiofind "PG.06"``

7. UART/I2C/CSI USAGE

1. UART Usage

	Jetpack 5.X	Jetpack 6.X/7.X
UART0	ttyTHS1	ttyTHS0
UART1	ttyTHS0	ttyTHS1

a. Install cutecom/minicom to use UART by below command.

- a) `sudo apt-get install cutecom`
- b) `sudo apt-get install minicom`

2. **I2C Usage** (I2C in carrier board is I2C@7), you can get the I2C device address by below command.

- a) `sudo i2cdetect -r -y 7`

3. **CSI Usage** (Example with dual IMX-219)

- a) Command: `sudo /opt/nvidia/jetson-io/jetson-io.py`

```
root@onx:/home/onx# sudo /opt/nvidia/jetson-io/jetson-io.py
```

- b) Select **Configure Jetson 22pin CSI Connector**.

```
===== Jetson Expansion Header Tool =====
Select one of the following:
Configure Jetson 40pin Header
Configure Jetson 22pin CSI Connector
Configure Jetson M.2 Key E Slot
Exit
```

- c) Select **Configure for compatible hardware**.

```
===== Jetson Expansion Header Tool =====  
  
3.3V ( 1) .. ( 2) i2c3  
i2c3 ( 3) .. ( 4) GND  
GND ( 7) .. ( 8) NA  
NA ( 9) .. (10) GND  
GND (13) .. (14) NA  
NA (15) .. (16) GND  
GND (19) .. (20) NA  
NA (21) .. (22) GND  
  
Jetson 22pin CSI Connector:  
Configure for compatible hardware  
Back
```

d) Select **Camera IMX219 Dual**.

```
===== Jetson Expansion Header Tool =====  
  
Select one of the following options:  
Camera IMX219 Dual  
Camera IMX219-A  
Camera IMX219-A and IMX477-C  
Camera IMX219-C  
Camera IMX477 Dual  
Camera IMX477-A  
Camera IMX477-A and IMX219-C  
Camera IMX477-C  
  
Back
```

e) Select **Save pin changes**.

```
===== Jetson Expansion Header Tool =====  
  
3.3V ( 1) .. ( 2) i2c3  
i2c3 ( 3) .. ( 4) GND  
GND ( 7) .. ( 8) NA  
NA ( 9) .. (10) GND  
GND (13) .. (14) NA  
NA (15) .. (16) GND  
GND (19) .. (20) NA  
NA (21) .. (22) GND  
  
Jetson 22pin CSI Connector:  
Save pin changes  
Discard pin changes
```

f) Select **Save and reboot to reconfigure pins**.

```
===== Jetson Expansion Header Tool =====  
  
Select one of the following:  
  
Configure Jetson 40pin Header  
Re-configure Jetson 22pin CSI Connector  
Configure Jetson M.2 Key E Slot  
Save and reboot to reconfigure pins  
Save and exit without rebooting  
Discard all pin changes  
Exit
```

g) Wait to re-boot and camera can be used.

WeAct Studio

CONTACT US

1. Github: <https://github.com/WeActStudio>
2. Gitee: <https://gitee.com/WeAct-TC>
3. Taobao: <https://weactstudio.taobao.com/>
4. Ali-express: Search WeAct in Ali-express

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